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Final Evaluation - Group Report

Basil Leaf Quality Classification

IT2011 - Artificial Intelligence and Machine Learning
Faculty of Computing | Year 2, Semester 1 (2026)

1. Introduction and Problem Statement

The task is to distinguish Healthy and Unhealthy basil leaf photographs using the assigned image dataset. The intended output is an experimental image label for an uploaded leaf photograph. It does not identify a specific disease, establish food safety or replace an agricultural assessment.

The group compares six classical supervised models: Logistic Regression, SVM, KNN, Decision Tree, Random Forest and Gradient Boosting. Each has a complete raw-image-to-prediction notebook and independently tuned settings. CNN is preserved as a separate supplementary experiment and excluded from this report's six-model ranking.

Measured outcome

The development-validation-selected model is Gradient Boosting. Its holdout accuracy is 98.67% and macro-F1 is 0.9864. This selection uses validation scores, not the best holdout score.

Scope and evidence

This report is generated from actual saved tuning tables, predictions, split manifests and metrics. It follows the nine sections requested in the supplied final-evaluation documentation. Historical progress work remains under progress/. No individual contribution or personal reflection is invented.

Assessment caveat

The fixed holdout was already evaluated in earlier project runs. It is excluded from the current search but must be described as a reused holdout, not a new external test. New-plant and new-location testing is still required.

2. Dataset Description / 3. Preprocessing and EDA

Assigned source: Leaves - India's Most Famous Basil Plant Leaves Quality Dataset, IEEE DataPort, DOI 10.21227/a4f6-4413. The local project records this attribution; the source landing page was not retrievable during report preparation.

Audit item	Observed value
Listed image files	1131
Unique valid images	1124
Exact duplicates removed	7
Rejected images	0
Perceptual groups	981
Development / holdout	899 / 225

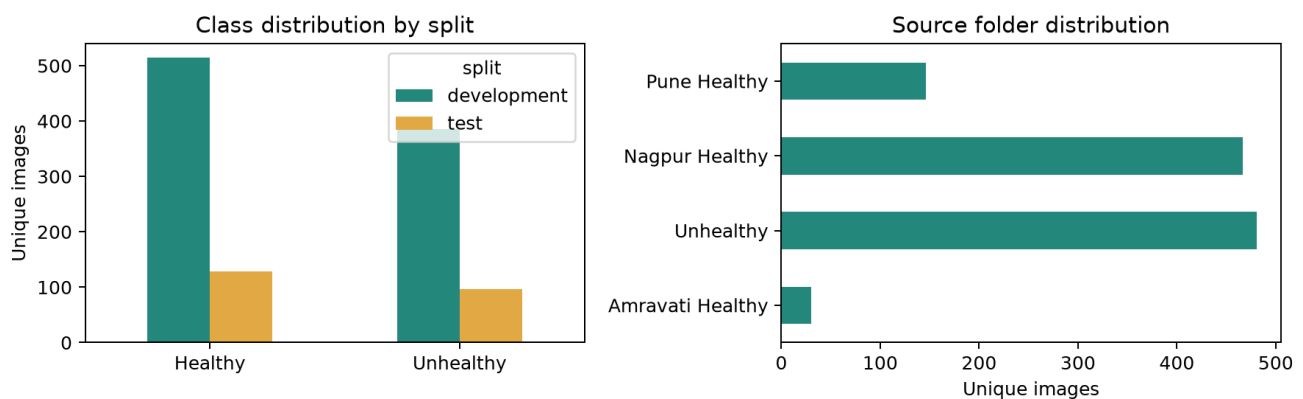


Figure 1. Class and source distributions after audit. Healthy examples come from three named regions, while unhealthy examples use one pooled folder. Health may therefore be confounded with source, background or capture conditions. Region-wise fairness cannot be estimated for unhealthy leaves from the available labels.

Images are decoded, EXIF-oriented, converted to RGB and validated. Duplicate decoded pixels are removed. Perceptual hashes within Hamming distance 4 are grouped before splitting. Grouping reduces similar-image leakage but does not establish independent plant identities.

No image-content outliers are deleted solely because they appear unusual; disease patterns may be informative. No missing numeric values require imputation. Historical progress PCA/scaled data are preserved but not reused for final fitting.

4. Model Design and Implementation

Complete reproducible pipeline

Raw labelled folders -> validation and duplicate audit -> grouped holdout split -> handcrafted features -> three grouped development folds -> fold-local scaling and feature selection -> parameter search -> refit on development images -> holdout metrics, plots and serialized model -> image prediction.

Feature representation

Every classical model receives the same 1,882-dimensional representation: 108 RGB/HSV histogram and summary features, 10 local binary pattern texture bins, and 1,764 HOG edge features. Color and texture use 128 x 128 images; HOG uses 64 x 64 grayscale. Fixed per-image feature extraction learns no dataset-level statistics.

Validation and tuning protocol

Seed 42 fixes an approximately 80/20 StratifiedGroupKFold holdout. On development images, identical three-fold grouped splits are reused across models. StandardScaler and SelectPercentile(f_classif) are fitted inside each training fold. Every parameter combination is tested with all features and with the top 25% features. Test labels are never used to fit these operations.

GridSearchCV selects maximum mean validation macro-F1 and refits on all development data. Accuracy and balanced accuracy are also logged for every candidate. Training scores and fold standard deviations expose overfitting and instability. Scores used for search selection are optimistic estimates, not nested-CV performance.

Model	Candidates	CV fits
Logistic Regression	8	24
SVM	8	24
KNN	12	36
Decision Tree	12	36
Random Forest	8	24
Gradient Boosting	8	24

Metrics and uncertainty

Macro-F1 is the mean of class F1 scores; each class contributes equally. Accuracy measures overall correctness, while balanced accuracy averages class recall. Per-class precision/recall and confusion matrices reveal false-healthy predictions. A 500-resample group bootstrap estimates holdout macro-F1 uncertainty; it does not account for training-set or model-selection uncertainty.

The following pages record every candidate variety and its validation outcome. Hyperparameters are limited to a practical CPU search, not an exhaustive global optimum.

4. Logistic Regression - individual pipeline

Regularized linear boundary; C controls complexity with high-dimensional features.

scikit-learn LogisticRegression: balanced class weights, max_iter=4000.

Winning configuration: {'model__C': 0.1, 'select__percentile': 100}

8 candidate varieties / 24 validation fits. Best CV macro-F1 0.9182 (fold SD 0.0170); holdout accuracy 92.44%; holdout macro-F1 0.9227. Search and refit time: 3.1 seconds.

ID	Parameter variety (model prefix omitted)	Train F1	CV F1	CV acc.
0	C=0.01, features %=25	0.978	0.906	0.908
1	C=0.01, features %=100	0.999	0.916	0.918
2	C=0.1, features %=25	0.999	0.912	0.914
3	C=0.1, features %=100	1.000	0.918	0.920
4	C=1, features %=25	1.000	0.910	0.912
5	C=1, features %=100	1.000	0.916	0.918
6	C=10, features %=25	1.000	0.908	0.910
7	C=10, features %=100	1.000	0.908	0.910

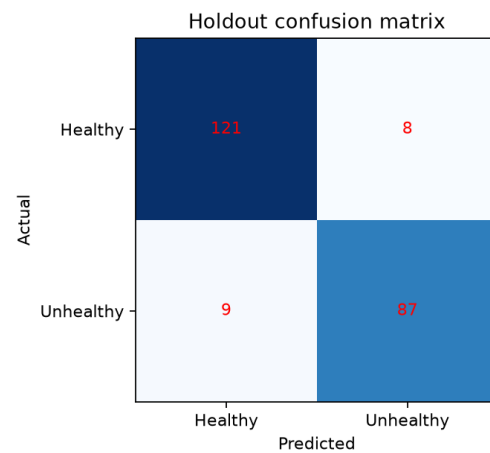
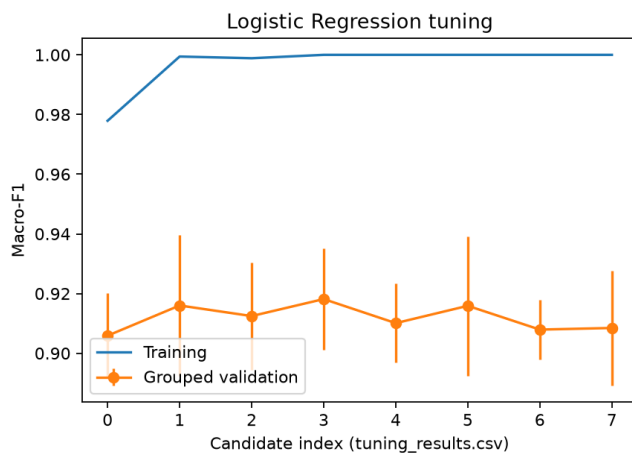


Figure: training/validation scores across candidate settings and final holdout confusion matrix. Candidate IDs match tuning_results.csv. Rows are actual classes; columns are predicted classes.

Conclusion: best reduced-feature CV F1 = 0.9125, versus 0.9182 with all features. The selected candidate has a training-minus-validation F1 gap of 0.0818. Retain the validation-selected setting; improvement on new images is not guaranteed. The holdout contains 9 unhealthy leaves predicted healthy.

4. SVM - individual pipeline

RBF kernel captures nonlinear boundaries; C and gamma control complexity and locality.

scikit-learn SVC: RBF kernel, balanced class weights.

Winning configuration: {'model__C': 10, 'model__gamma': 'scale', 'select__percentile': 25}

8 candidate varieties / 24 validation fits. Best CV macro-F1 0.9463 (fold SD 0.0113); holdout accuracy 97.33%; holdout macro-F1 0.9726. Search and refit time: 37.0 seconds.

ID	Parameter variety (model prefix omitted)	Train F1	CV F1	CV acc.
0	C=1, gamma=scale, features %=25	0.998	0.942	0.943
1	C=1, gamma=scale, features %=100	0.999	0.939	0.941
2	C=1, gamma=0.001, features %=25	0.989	0.932	0.933
3	C=1, gamma=0.001, features %=100	1.000	0.929	0.932
4	C=10, gamma=scale, features %=25	1.000	0.946	0.948
5	C=10, gamma=scale, features %=100	1.000	0.943	0.944
6	C=10, gamma=0.001, features %=25	1.000	0.945	0.947
7	C=10, gamma=0.001, features %=100	1.000	0.933	0.936

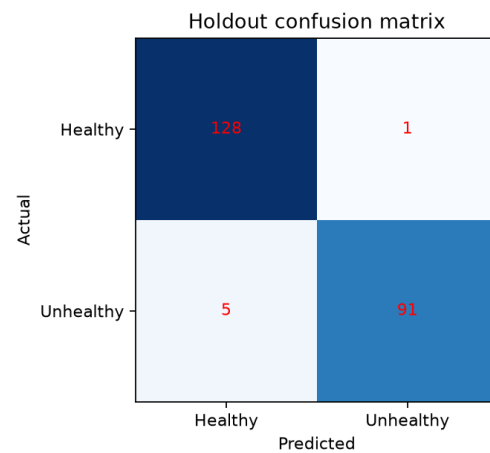
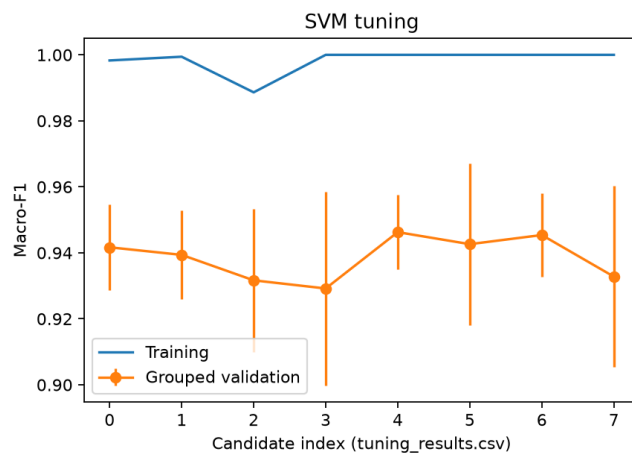


Figure: training/validation scores across candidate settings and final holdout confusion matrix. Candidate IDs match tuning_results.csv. Rows are actual classes; columns are predicted classes.

Conclusion: best reduced-feature CV F1 = 0.9463, versus 0.9426 with all features. The selected candidate has a training-minus-validation F1 gap of 0.0537. Retain the validation-selected setting; improvement on new images is not guaranteed. The holdout contains 5 unhealthy leaves predicted healthy.

4. KNN - individual pipeline

Local similarity baseline; neighbors and distance weights control smoothing.

scikit-learn KNeighborsClassifier: Minkowski distance, p=2 (Euclidean).

Winning configuration: {'model__n_neighbors': 3, 'model__weights': 'distance', 'select__percentile': 25}

12 candidate varieties / 36 validation fits. Best CV macro-F1 0.9272 (fold SD 0.0214); holdout accuracy 94.22%; holdout macro-F1 0.9412. Search and refit time: 10.9 seconds.

ID	Parameter variety (model prefix omitted)	Train F1	CV F1	CV acc.
0	n_neighbors=3, weights=uniform, features %=25	0.981	0.926	0.928
1	n_neighbors=3, weights=uniform, features %=100	0.970	0.904	0.907
2	n_neighbors=3, weights=distance, features %=25	1.000	0.927	0.929
3	n_neighbors=3, weights=distance, features %=100	1.000	0.904	0.907
4	n_neighbors=5, weights=uniform, features %=25	0.965	0.909	0.911
5	n_neighbors=5, weights=uniform, features %=100	0.951	0.887	0.891
6	n_neighbors=5, weights=distance, features %=25	1.000	0.910	0.912
7	n_neighbors=5, weights=distance, features %=100	1.000	0.889	0.893
8	n_neighbors=11, weights=uniform, features %=25	0.938	0.890	0.893
9	n_neighbors=11, weights=uniform, features %=100	0.910	0.865	0.872
10	n_neighbors=11, weights=distance, features %=25	1.000	0.891	0.894
11	n_neighbors=11, weights=distance, features %=100	1.000	0.866	0.873

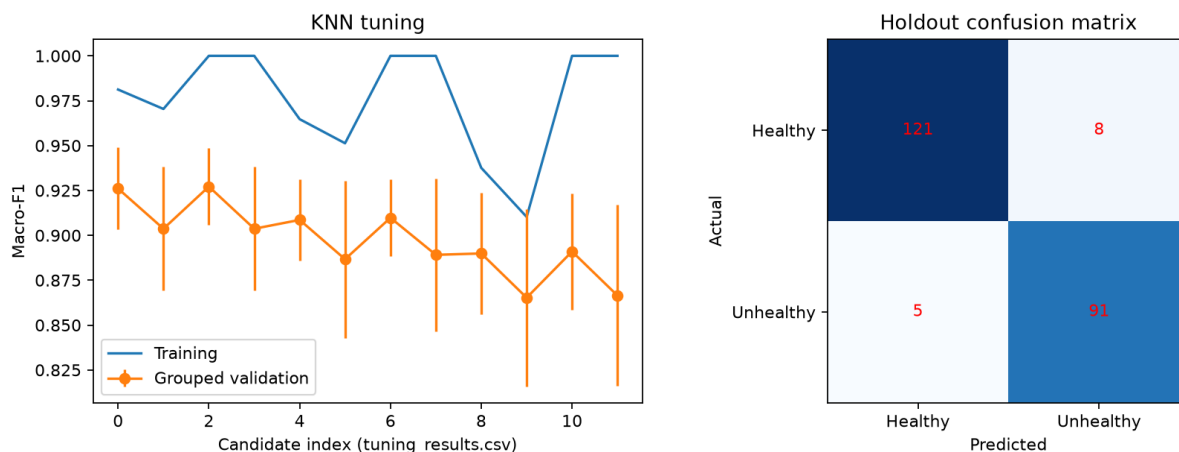


Figure: training/validation scores across candidate settings and final holdout confusion matrix. Candidate IDs match tuning_results.csv. Rows are actual classes; columns are predicted classes.

Conclusion: best reduced-feature CV F1 = 0.9272, versus 0.9037 with all features. The selected candidate has a training-minus-validation F1 gap of 0.0728. Retain the validation-selected setting; improvement on new images is not guaranteed. The holdout contains 5 unhealthy leaves predicted healthy.

4. Decision Tree - individual pipeline

Interpretable nonlinear rules; depth and leaf size limit overfitting.

scikit-learn DecisionTreeClassifier: Gini criterion, balanced class weights.

Winning configuration: {'model__max_depth': None, 'model__min_samples_leaf': 2, 'select__percentile': 25}

12 candidate varieties / 36 validation fits. Best CV macro-F1 0.9151 (fold SD 0.0272); holdout accuracy 92.00%; holdout macro-F1 0.9173. Search and refit time: 23.9 seconds.

ID	Parameter variety (model prefix omitted)	Train F1	CV F1	CV acc.
0	max_depth=3, min_samples_leaf=2, features %=25	0.956	0.900	0.902
1	max_depth=3, min_samples_leaf=2, features %=100	0.959	0.901	0.903
2	max_depth=3, min_samples_leaf=5, features %=25	0.954	0.893	0.895
3	max_depth=3, min_samples_leaf=5, features %=100	0.956	0.897	0.899
4	max_depth=6, min_samples_leaf=2, features %=25	0.985	0.913	0.914
5	max_depth=6, min_samples_leaf=2, features %=100	0.993	0.897	0.899
6	max_depth=6, min_samples_leaf=5, features %=25	0.971	0.890	0.892
7	max_depth=6, min_samples_leaf=5, features %=100	0.981	0.891	0.893
8	max_depth=None, min_samples_leaf=2, features %=25	0.991	0.915	0.917
9	max_depth=None, min_samples_leaf=2, features %=100	0.997	0.896	0.898
10	max_depth=None, min_samples_leaf=5, features %=25	0.971	0.888	0.890
11	max_depth=None, min_samples_leaf=5, features %=100	0.981	0.891	0.893

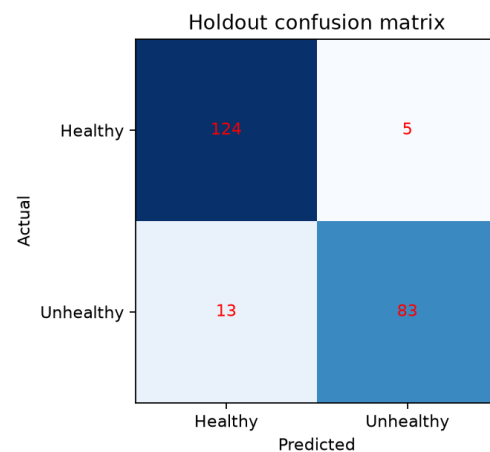
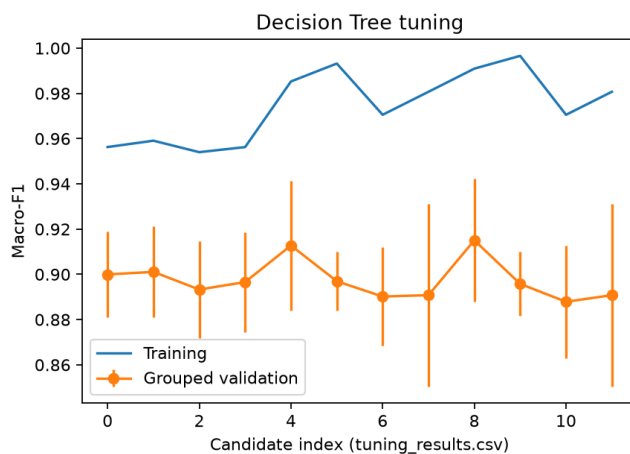


Figure: training/validation scores across candidate settings and final holdout confusion matrix. Candidate IDs match tuning_results.csv. Rows are actual classes; columns are predicted classes.

Conclusion: best reduced-feature CV F1 = 0.9151, versus 0.9011 with all features. The selected candidate has a training-minus-validation F1 gap of 0.0759. Retain the validation-selected setting; improvement on new images is not guaranteed. The holdout contains 13 unhealthy leaves predicted healthy.

4. Random Forest - individual pipeline

Averaged randomized trees reduce variance; leaf size and feature sampling regulate complexity.

scikit-learn RandomForestClassifier: 200 trees, Gini criterion, balanced class weights.

Winning configuration: {'model__max_features': 'sqrt', 'model__min_samples_leaf': 1, 'select__percentile': 100}

8 candidate varieties / 24 validation fits. Best CV macro-F1 0.9657 (fold SD 0.0049); holdout accuracy 97.78%; holdout macro-F1 0.9771. Search and refit time: 196.6 seconds.

ID	Parameter variety (model prefix omitted)	Train F1	CV F1	CV acc.
0	max_features=sqrt, min_samples_leaf=1, features %=25	1.000	0.965	0.966
1	max_features=sqrt, min_samples_leaf=1, features %=100	1.000	0.966	0.967
2	max_features=sqrt, min_samples_leaf=3, features %=25	1.000	0.958	0.959
3	max_features=sqrt, min_samples_leaf=3, features %=100	1.000	0.960	0.961
4	max_features=0.3, min_samples_leaf=1, features %=25	1.000	0.962	0.963
5	max_features=0.3, min_samples_leaf=1, features %=100	1.000	0.957	0.958
6	max_features=0.3, min_samples_leaf=3, features %=25	1.000	0.961	0.962
7	max_features=0.3, min_samples_leaf=3, features %=100	1.000	0.957	0.958

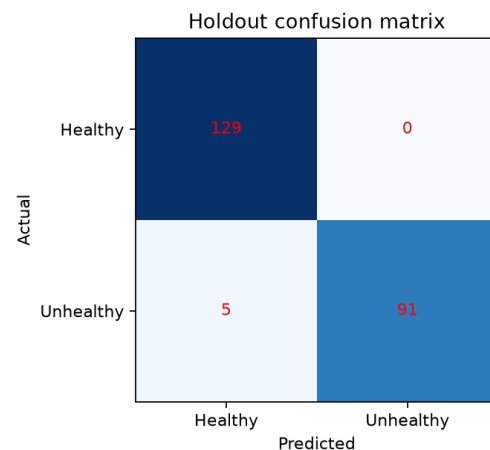
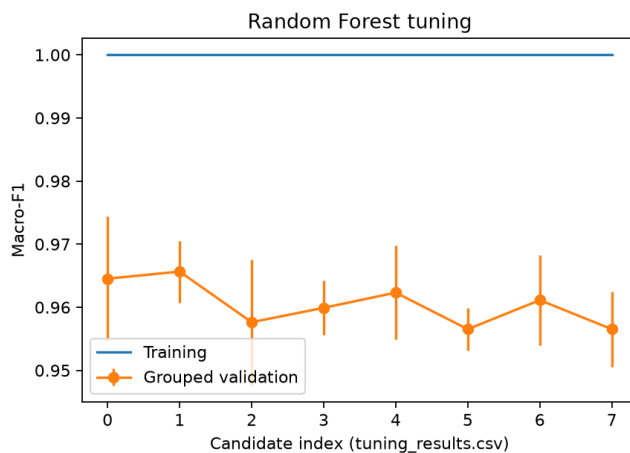


Figure: training/validation scores across candidate settings and final holdout confusion matrix. Candidate IDs match tuning_results.csv. Rows are actual classes; columns are predicted classes.

Conclusion: best reduced-feature CV F1 = 0.9645, versus 0.9657 with all features. The selected candidate has a training-minus-validation F1 gap of 0.0343. Retain the validation-selected setting; improvement on new images is not guaranteed. The holdout contains 5 unhealthy leaves predicted healthy.

4. Gradient Boosting - individual pipeline

Sequential trees correct residual errors; learning rate and depth regulate complexity.

scikit-learn GradientBoostingClassifier: 100 trees, max_features=sqrt, log-loss.

Winning configuration: {'model__learning_rate': 0.1, 'model__max_depth': 3, 'select__percentile': 25}

8 candidate varieties / 24 validation fits. Best CV macro-F1 0.9727 (fold SD 0.0097); holdout accuracy 98.67%; holdout macro-F1 0.9864. Search and refit time: 30.2 seconds.

ID	Parameter variety (model prefix omitted)	Train F1	CV F1	CV acc.
0	learning_rate=0.05, max_depth=2, features %=25	0.986	0.954	0.955
1	learning_rate=0.05, max_depth=2, features %=100	0.987	0.951	0.952
2	learning_rate=0.05, max_depth=3, features %=25	1.000	0.965	0.965
3	learning_rate=0.05, max_depth=3, features %=100	1.000	0.962	0.963
4	learning_rate=0.1, max_depth=2, features %=25	1.000	0.964	0.964
5	learning_rate=0.1, max_depth=2, features %=100	1.000	0.962	0.963
6	learning_rate=0.1, max_depth=3, features %=25	1.000	0.973	0.973
7	learning_rate=0.1, max_depth=3, features %=100	1.000	0.968	0.969

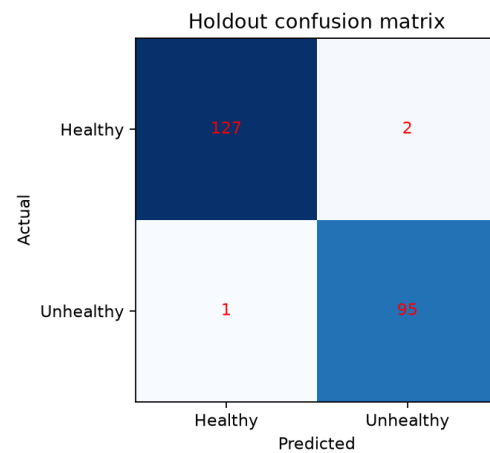
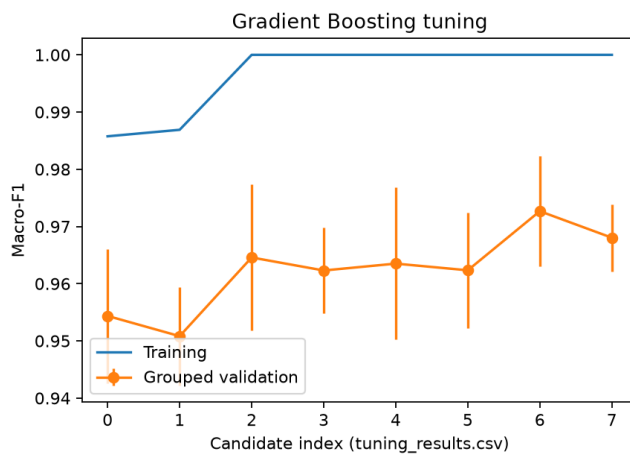


Figure: training/validation scores across candidate settings and final holdout confusion matrix. Candidate IDs match tuning_results.csv. Rows are actual classes; columns are predicted classes.

Conclusion: best reduced-feature CV F1 = 0.9727, versus 0.9680 with all features. The selected candidate has a training-minus-validation F1 gap of 0.0273. Retain the validation-selected setting; improvement on new images is not guaranteed. The holdout contains 1 unhealthy leaves predicted healthy.

5. Evaluation and Comparison - exactly six models

Models are ordered by development CV macro-F1. CNN is excluded. Each holdout column uses identical image paths and labels.

Model	CV F1	Test acc.	Test F1	95% group-bootstrap F1
Gradient Boosting	0.9727	98.67%	0.9864	0.9682 - 1.0000
Random Forest	0.9657	97.78%	0.9771	0.9514 - 0.9953
SVM	0.9463	97.33%	0.9726	0.9496 - 0.9907
KNN	0.9272	94.22%	0.9412	0.8979 - 0.9771
Logistic Regression	0.9182	92.44%	0.9227	0.8843 - 0.9562
Decision Tree	0.9151	92.00%	0.9173	0.8572 - 0.9639

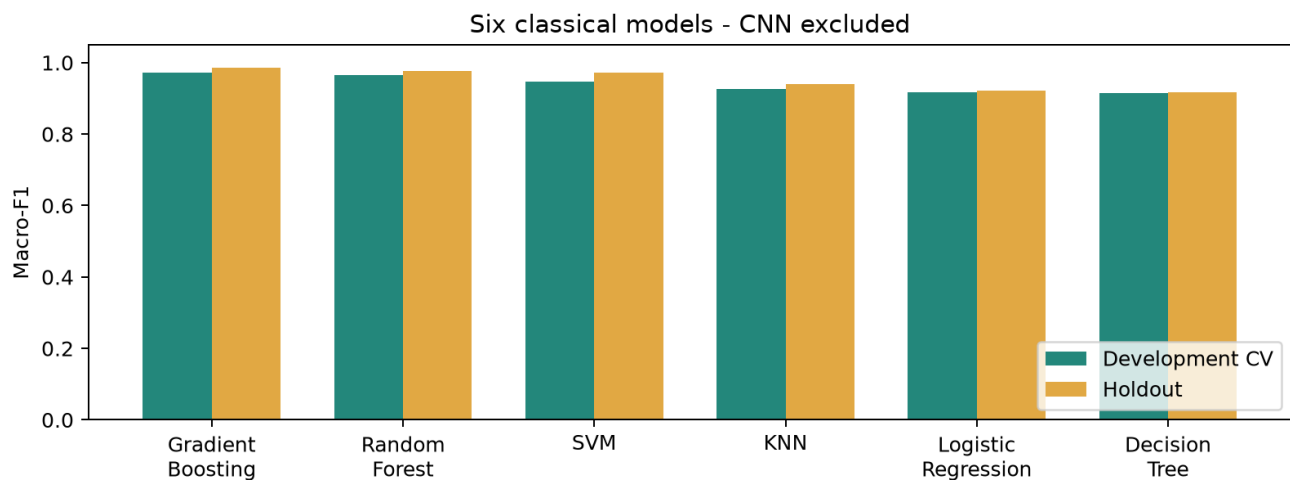


Figure 2. Validation and holdout macro-F1. Different rankings across the two sets are expected with limited data; the holdout ranking does not change the selected model.

Gradient Boosting had the highest mean grouped 3-fold development macro-F1 (0.9727). Macro-F1 weights each class equally. Exact score ties are broken by mean classifier fitting time, then model name. The holdout was excluded from selection.

The majority-class rule would achieve 57.33% holdout accuracy. This is a reference calculation, not a seventh trained model.

Practical trade-offs

Distance-based methods depend on feature scaling and dimensionality. Trees offer nonlinear decisions but can overfit. Ensembles reduce some instability at additional fitting cost. Timing here is descriptive because training workloads shared this computer; classifier latency excludes image decoding and feature extraction. No statistical superiority claim follows from point estimates or overlapping intervals.

6. Ethics / 7. AI Transparency / 8. Reflections

6. Ethical Considerations and Bias Mitigation

Dataset labels describe image quality/health categories rather than verified disease diagnosis. A false-healthy result can mislead users; the interface exposes this limitation. Local uploads are decoded in memory and are not retained or used for training. Respect the original dataset terms before redistributing images.

Class-weighting is used in Logistic Regression, SVM, Decision Tree, Random Forest and CNN. Macro-F1 and class recall avoid reporting majority-class accuracy alone. Grouped splits reduce duplicate leakage. These measures do not remove source/background bias, unknown plant identity, imperfect labels or absent external validation.

7. AI Tool Usage Declaration and Transparency

OpenAI Codex assisted with inspecting the supplied briefs, restructuring code and notebooks, implementing tuning, creating visualizations, checking artifacts and drafting this report. Metrics and figures were computed by running code on the local dataset; no performance values were invented. The group must review and understand the generated work and declare AI use according to course policy. AI assistance is not evidence of individual student contribution.

8. Reflections and Lessons Learned

Technical lessons from the implementation: identical splits are necessary for comparisons; preprocessing must be fitted inside validation folds; tuning must record failed or inferior choices as well as winners; deployment must use the exact training feature implementation. Search variety and model suitability matter more than listing many algorithms.

A large training-validation gap indicates overfitting; reducing dimensions may help distance methods but can remove useful interactions. Search scores cannot justify claiming general improvement. A reused holdout and region-label confounding remain major limitations despite high point scores.

These are implementation reflections, not attributed personal experiences. Before the viva, each member should explain their own notebook, parameter varieties, output files, errors and improvement proposals. The supplied requirement explicitly assesses individual demonstration.

Next research steps: collect a prospective external set with plant IDs, healthy/unhealthy examples from each region and varied backgrounds; consider segmentation and calibrated confidence using development-only validation. These are proposed improvements, not completed experiments.

9. References and Reproduction

[1] Supplied course brief: final/Final Evaluation - Documentation.pdf. Nine report sections, maximum 15 pages, group-ID title.

[2] Supplied course brief: final/Final Evaluation - Implementation.pdf. Individual model/tuning evidence and comparison of six optimum individual models.

[3] Assigned dataset attribution recorded in project: IEEE DataPort, Leaves - India's Most Famous Basil Plant Leaves Quality Dataset. DOI: 10.21227/a4f6-4413. Source landing page unavailable during this review.

[4] scikit-learn official documentation, GridSearchCV.

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html (accessed 2026-09-27).

[5] scikit-learn official documentation, StratifiedGroupKFold.

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.StratifiedGroupKFold.html (accessed 2026-09-27).

Reproduce from zero

Install requirements.txt into a Python virtual environment. Place the four assigned labelled image folders under data/raw. Run python _run_all_training.py for all six models, or add --model MODEL_KEY for an individual model. Run python CNN/train_cnn.py for the separate CNN experiment. Generate both PDFs with python build_report.py --group-id abc.

Predict an image with python predict.py IMAGE --model MODEL_KEY, or run python app.py for the local selected-model demonstration. Each member notebook executes audit, feature extraction, tuning, evaluation, export and reload/inference.

Evidence locations

parts/<model>/outputs/: tuning_results.csv, preprocessing_comparison.csv, search_space.json, model bundle, metrics, predictions, split manifest and evaluation.png.

outputs/: shared audit, environment versions, selected deployment model and six_model_comparison.csv.

CNN/: separate model, trial history, checkpoint, metrics and predictions.

progress/: preserved prior deliverables.

final/: preserved assignment briefs.

Dataset fingerprint: 45649c31a10fc7497fd7e5875b8adcf043ed607a184cb90742f06882bf040e1d

Runtime: 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)]